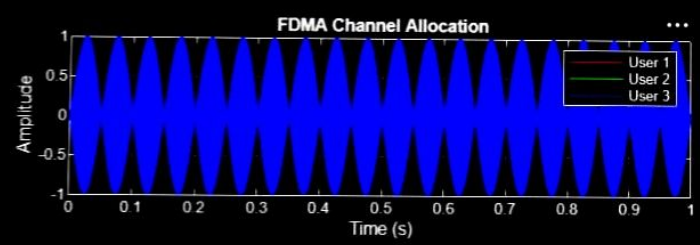
```
46      grid on;
47
48      xlabel('Users');
49      ylabel('Channel Bandwidth (kHz)');
50
51      title('Channel Bandwidth per User');
52
53      xticks(1:Users);
54      xticklabels({'User 1','User 2','User 3'});
55
56      %% Display Results
57      fprintf('===== FDMA CHANNEL ALLOCATION =====\n');
58      fprintf('Total Bandwidth       = %d kHz\n',TBW);
59      fprintf('Number of Users       = %d\n',Users);
60      fprintf('Bandwidth/User        = %d kHz\n',CBW);
61
62      for i = 1:Users
63          fprintf('User %d Carrier Frequency = %d Hz\n', ...
64              i,fc(i));
```



```m
clc;
clear;
close all;

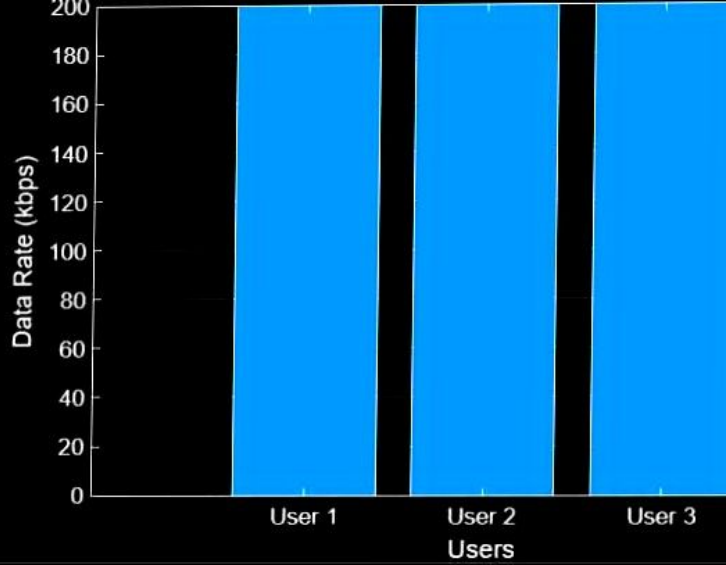
BW = 200;               % Channel Bandwidth (kHz)
Users = 3;              % Number of Users
M = 2;                  % BPSK Modulation

DataRate = BW * log2(M);   % Data Rate (kbps)
Rate = DataRate * ones(1, Users);

disp(' User   Data Rate (kbps)')
disp([(1:Users)' Rate'])

figure;
bar(1:Users, Rate);

xlabel('Users');
ylabel('Data Rate (kbps)');
```



```
Users = 1:3;

SNR = 10*log10(Ps./Pn);          % SNR in dB

disp('User   Noise Power(W)   SNR(dB)')
disp([Users' Pn' SNR'])

figure;

bar(Users,SNR);

xlabel('Users');
ylabel('SNR (dB)');
title('Signal-to-Noise Ratio of FDMA Users');

grid on;

xticks(Users);
xticklabels({'User 1','User 2','User 3'});
```

